

A Volumetric Comparison of the Vestibular Nuclei in Primates

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Abstract. The volumes of each of the four vestibular nuclei, superior, lateral, medial and descending, were measured in 80 brains from 2 species of Scandentia, 18 species of prosimians, and 26 species of anthropoids. Size indices were calculated by comparing species-specific points to the nucleus volume-body weight allometry in prosimians, where the average prosimian was set at 1.00. The indices range from 1.78 in *Saimiri* to 0.48 in *Gorilla*, and the distributions by families overlap partially or completely. The observed trend in size indices is independent of changes in the neocortex and the ventral pons; average indices are 1.35 in New World monkeys, 1.20 in Old World monkeys, 0.74 in apes, 0.82 in man. Among prosimians, *Galago*, *Galagoides* and *Tarsius* (leaping locomotion) show significantly higher indices than *Nycticebus*, *Loris* and *Perodicticus* (slow movement without leaping). The lateral vestibular nuclear indices in Pongidae and man are extremely low, about half of those of the average prosimians. Correlation coefficients of size indices between the vestibular nuclei and other motor nuclei, such as the cerebellar nuclei, ventral pons and striatum, are analysed. The ratio of the vestibular nuclear volumes to the total brain volumes and the distribution of percentages of each vestibular nuclear volume to the total complex are also obtained.

Introduction

Among the vast amount of brain research, progress on the morphology and function of the vestibular system is noteworthy [Kornhuber, 1974; Granit and Pompeiano, 1979]. However, research from a comparative and evolutionary perspective is still insufficient, especially in primates.

The structural organization of a nucleus is closely related to its function and behavioral importance [Masterton et al., 1976; King, 1978]. In general, enlargement of a nucleus is considered to reflect an increase in the number of neurons, in the size of the dendritic arborization and in the numbers of the synaptic terminals and other complexities in its neuronal structures. Therefore, allometric

comparisons among various species in an order or a family using the size of a nucleus and body size are very interesting. The phylogenetic development of a nucleus and its adaptation to species-specific behaviour can be investigated through such comparisons.

From this point of view, Baron [1977] related quantitative differences between vestibular nuclei in many species of bats to their flight behavior. He concluded that there is a close correspondence between the development of allometric size of the vestibular nuclei and flight behavior, and hypothesized that the development of the vestibular nuclei indicates a specialization in locomotion, whereas the degree of neocorticalization may indicate a level of intelligence or cognition.

An early comparative study of the vestibular nuclei in primates noted the vestibular nuclei of *Tarsius*, to be especially large [Tilney, 1927]. The topography and cytoarchitecture of the vestibular complex have also been reported for a number of mammals: rabbits [Meessen and Olszewski, 1949], cats [Brodal and Pompeiano, 1957], opossums [Henkel and Martin, 1977], bushbabies [Rubertone and Haines, 1982], monkeys [Brodal, 1984] and man [Sadjadpour and Brodal, 1968]. Gerhard and Olszewski [1969] compared species-specific characteristics of the vestibular nuclei among some primate species. Recently, Matano et al. [1985a, c] compared the allometric size of the ventral pons and the cerebellar nuclei of various primates based on the prosimian level, and discussed their relation to the forms of locomotion. The present paper compares the volumes of the vestibular nuclei among primates and Scandentia, discussing the relation to size of the labyrinthine receptors, the cerebellar motor system and the forms of locomotion.

The oculomotor system is also functionally related to the vestibular system, but it will be discussed in another paper. The data of Stephan et al. [1981] have been incorporated in the present study.

Materials and Methods

The volumes of the vestibular nuclei were measured in 80 individuals from 46 species: 2 species of Scandentia and 44 primate species (18 prosimians and 26 anthropoids including man). The vestibular nuclear complex was divided into 4 main nuclei, i.e., the superior, medial, lateral and descending nucleus. Several small cell groups of the vestibular complex described by Brodal and Pompeiano [1957] and Meessen and Olszewski [1949] were ignored (nucleus parasolarius, group y) or included in one of the 4 main nuclei (interstitial nucleus of the vestibular nerve, group l in the lateral vestibular nucleus; group f, group x, group z in the descending vestibular nucleus; and nucleus supravestibularis in the medial vestibular nucleus [Baron, 1977]).

As a rule, brains were fixed in Bouin's fluid (a few in 10% formalin) and embedded in paraffin. Frontal serial sections were stained for cells with cresyl-violet and for fibers by the Heidenhain-Woelcke method. Five to 8 sections of each vestibular nucleus were taken at equal intervals in serial sections, and projected directly on photographic paper (18 × 24 cm); the border of each nucleus was delineated, which was then cut out and weighed. Distances between the histological sections were calculated, and the volume of each nucleus was derived from the formula:

$$V = AP \times WS \times D/M^2,$$

where V = volume in mm³; AP = average area of the photographic paper in mm²/mg; WS = weight of the photographic paper of each vestibular nucleus in mg; D = distance of measured sections in mm; M² = square of linear magnification. The calculated volumes were converted to estimated fresh volumes according to Stephan et al. [1981]. The obtained results are listed in table I.

In this paper allometric methods of comparison were used. Size indices, which evaluate the size differences of each vestibular nucleus in isoponderal species, were determined (table II and IV). The volume of each nucleus and complex was plotted against body

Table I. Volumes (mm³) of the vestibular nuclei in primates

	n	Body weight g	Vestibular nuclei				
			superior	lateral	medial	descending	complex
Scandentia							
<i>Tupaia glis</i>	2	170	2.16	2.07	3.27	2.62	10.1
<i>Urogale everetti</i>	1	275	1.91	2.33	4.68	3.59	12.5
Primates, prosimians							
<i>Microcebus murinus</i>	2	54	0.679	0.760	1.50	1.08	4.02
<i>Cheirogaleus major</i>	2	450	1.86	1.78	4.18	3.27	11.1
<i>C. medius</i>	1	177	1.36	1.00	2.03	1.61	6.00
<i>Lemur albifrons</i>	2	1,400	6.22	5.90	9.75	9.27	31.1
<i>Varecia variegata</i>	1	3,000	6.92	7.86	14.8	13.8	43.4
<i>Lepilemur ruficaudatus</i>	1	915	2.11	2.38	5.92	4.47	14.9
<i>Avahi laniger laniger</i>	1	1,285	4.27	5.02	9.21	5.19	23.7
<i>A. l. occidentalis</i>	1	860	3.10	3.32	6.86	4.81	18.1
<i>Propithecus verreauxi</i>	2	3,480	8.18	7.37	15.3	12.4	43.3
<i>Indri indri</i>	2	6,250	7.80	9.51	22.2	11.8	51.3
<i>Daubentonia madagascariensis</i>	1	2,800	8.45	8.13	18.0	12.7	47.3
<i>Loris tardigradus</i>	2	322	1.61	1.22	3.55	2.08	8.46
<i>Nycticebus coucang</i>	2	800	3.03	3.05	7.32	5.14	18.5
<i>Perodicticus potto</i>	2	1,150	2.86	2.66	7.87	3.88	17.3
<i>Galago senegalensis</i>	2	186	1.36	1.43	3.42	2.93	9.14
<i>Otolemur crassicaudatus</i>	2	850	2.85	2.87	7.16	7.01	19.9
<i>Galagoides demidovii</i>	2	81	1.02	1.01	2.72	1.90	6.65
<i>Tarsius spp.</i>	4	125	0.759	1.16	3.36	2.41	7.69
Primates, simians							
<i>Callithrix jacchus</i>	2	280	2.69	4.07	5.80	2.60	15.2
<i>Cebuella pygmaea</i>	2	140	2.10	2.94	4.11	1.96	11.1
<i>Saguinus midas</i>	2	340	3.21	3.28	5.70	3.60	15.8
<i>S. oedipus</i>	2	380	2.98	3.63	7.25	4.49	18.4
<i>Cebus albifrons</i>	1	3,100	11.2	7.45	19.2	12.3	50.2
<i>Aotus trivirgatus</i>	2	830	6.07	5.14	11.1	5.81	28.1
<i>Callicebus moloch</i>	2	900	6.17	5.29	13.1	8.52	33.1
<i>Saimiri sciureus</i>	2	660	4.43	6.26	11.4	7.31	29.4
<i>Pithecia monachus</i>	2	1,500	8.07	8.58	12.0	10.1	38.8
<i>Alouatta seniculus</i>	2	6,400	9.09	9.27	16.3	11.2	45.9
<i>Ateles geoffroyi</i>	1	8,000	7.20	6.81	23.8	11.9	49.7
<i>Lagothrix lagotricha</i>	2	5,200	9.07	6.94	23.0	14.1	53.1
<i>Macaca mulatta</i>	2	7,800	16.4	16.9	33.7	15.5	82.5
<i>Cercopithecus albigena</i>	2	7,900	16.2	10.0	35.6	25.2	87.0
<i>Papio anubis</i>	2	25,000	14.4	10.5	48.1	33.9	107
<i>Cercopithecus ascanius</i>	2	3,400	8.21	5.88	20.8	17.2	52.1
<i>C. mitis</i>	1	6,300	15.0	10.5	30.7	24.3	80.5
<i>Miopithecus talapoin</i>	2	1,200	7.61	7.25	15.7	9.24	39.8
<i>Erythrocebus patas</i>	2	7,800	15.2	17.9	40.3	15.6	89.0
<i>Colobus badius</i>	2	7,000	8.68	10.0	24.1	14.5	57.3
<i>Pygathrix nemaeus</i>	1	7,500	14.0	12.2	18.1	18.2	62.5
<i>Nasalis larvatus</i>	1	14,000	12.6	14.8	33.6	25.2	86.2
<i>Hylobates lar</i>	1	5,700	14.9	5.53	24.5	14.5	59.4
<i>Pan troglodytes</i>	1	46,000	23.5	9.65	54.7	29.8	118
<i>Gorilla gorilla</i>	2	105,000	34.7	19.5	58.4	23.6	136
<i>Homo sapiens sapiens</i>	2	65,000	35.8	22.4	72.8	45.9	177

Table II. Size indices for the slope of 0.56

	n	Vestibular nuclei				
		superior ¹	lateral ²	medial ³	descending ⁴	complex ⁵
Scandentia						
<i>Tupaia glis</i>	2	1.68	1.58	1.11	1.20	1.31
<i>Urogale everetti</i>	1	1.14	1.36	1.21	1.26	1.24
Primates, prosimians						
<i>Microcebus murinus</i>	2	1.01	1.11	0.96	0.94	0.99
<i>Cheirogaleus major</i>	2	0.84	0.79	0.82	0.87	0.83
<i>C. medius</i>	1	1.04	0.75	0.67	0.72	0.76
<i>Lemur albifrons</i>	2	1.49	1.39	1.01	1.30	1.24
<i>Varecia variegata</i>	1	1.08	1.21	1.00	1.27	1.12
<i>Lepilemur ruficaudatus</i>	1	0.64	0.71	0.78	0.80	0.75
<i>Avahi laniger laniger</i>	1	1.08	1.24	1.00	0.77	0.99
<i>A. l. occidentalis</i>	1	0.98	1.03	0.94	0.89	0.94
<i>Propithecus verreauxi</i>	2	1.18	1.04	0.95	1.05	1.03
<i>Indri indri</i>	2	0.81	0.97	1.00	0.72	0.88
<i>Daubentonia madagascariensis</i>	1	1.38	1.30	1.27	1.21	1.27
<i>Loris tardigradus</i>	2	0.88	0.65	0.84	0.67	0.76
<i>Nycticebus coucang</i>	2	0.99	0.98	1.04	0.99	1.01
<i>Perodicticus potto</i>	2	0.77	0.70	0.91	0.61	0.77
<i>Galago senegalensis</i>	2	1.01	1.04	1.10	1.28	1.12
<i>Otolemur crassicaudatus</i>	2	0.91	0.89	0.98	1.30	1.05
<i>Galagoides demidovii</i>	2	1.20	1.17	1.39	1.32	1.30
<i>Tarsius spp.</i>	4	0.70	1.06	1.35	1.31	1.18
Primates, simians						
<i>Callithrix jacchus</i>	2	1.59	2.36	1.48	0.90	1.48
<i>Cebuella pygmaea</i>	2	1.83	2.51	1.55	1.01	1.61
<i>Saguinus midas</i>	2	1.70	1.70	1.31	1.12	1.39
<i>S. oedipus</i>	2	1.49	1.77	1.56	1.31	1.51
<i>Cebus albifrons</i>	1	1.72	1.12	1.28	1.11	1.28
<i>Aotus trivirgatus</i>	2	1.95	1.62	1.54	1.10	1.50
<i>Callicebus moloch</i>	2	1.90	1.59	1.74	1.53	1.68
<i>Saimiri sciureus</i>	2	1.62	2.24	1.81	1.57	1.78
<i>Pithecia monachus</i>	2	1.86	1.94	1.20	1.37	1.48
<i>Alouatta seniculus</i>	2	0.93	0.93	0.72	0.67	0.78
<i>Ateles geoffroyi</i>	1	0.65	0.60	0.93	0.63	0.74
<i>Lagothrix lagotricha</i>	2	1.04	0.78	1.14	0.95	1.01
<i>Macaca mulatta</i>	2	1.50	1.52	1.34	0.83	1.25
<i>Cercocebus albigena</i>	2	1.48	0.90	1.40	1.35	1.31
<i>Papio anubis</i>	2	0.69	0.49	0.99	0.95	0.85
<i>Cercopithecus ascanius</i>	2	1.20	0.84	1.31	1.47	1.26
<i>C. mitis</i>	1	1.55	1.06	1.37	1.47	1.38
<i>Miopithecus talapoin</i>	2	1.99	1.86	1.78	1.42	1.72
<i>Erythrocebus patas</i>	2	1.39	1.61	1.60	0.84	1.35
<i>Colobus badius</i>	2	0.85	0.96	1.02	0.83	0.93
<i>Pygathrix nemaeus</i>	1	1.31	1.12	0.73	1.00	0.97
<i>Nasalis larvatus</i>	1	0.83	0.96	0.96	0.98	0.94
<i>Hylobates lar</i>	1	1.62	0.59	1.16	0.93	1.07
<i>Pan troglodytes</i>	1	0.80	0.32	0.80	0.59	0.66
<i>Gorilla gorilla</i>	2	0.74	0.41	0.54	0.30	0.48
<i>Homo sapiens sapiens</i>	2	1.00	0.61	0.88	0.75	0.82

¹ Superior vestibular nucleus $\log y = 0.56 \log x - 1.142$.² Lateral vestibular nucleus $\log y = 0.56 \log x - 1.133$.³ Medial vestibular nucleus $\log y = 0.56 \log x - 0.778$.⁴ Descending vestibular nucleus $\log y = 0.56 \log x - 0.910$.⁵ Complex vestibular nucleus $\log y = 0.56 \log x - 0.361$.

Table III. Ratios of vestibular nuclei volumes

	n	Vestibular nuclei				
		superior %	lateral %	medial %	descending %	complex to total brain × 100,000
Scandentia						
<i>Tupaia glis</i>	2	21.4	20.5	32.3	25.8	337
<i>Urogale everetti</i>	1	15.3	18.6	37.4	28.7	313
Primates, Prosimians						
<i>Microcebus murinus</i>	2	16.9	18.9	37.3	26.9	239
<i>Cheirogaleus major</i>	2	16.8	16.0	37.7	29.5	174
<i>C. medius</i>	1	22.7	16.7	33.8	26.8	203
<i>Lemur albifrons</i>	2	20.0	19.0	31.3	29.7	141
<i>Varecia variegata</i>	1	15.9	18.2	34.1	31.8	146
<i>Lepilemur ruficaudatus</i>	1	14.2	16.1	39.7	30.0	208
<i>Avahi laniger laniger</i>	1	18.0	21.2	38.9	21.9	242
<i>A. l. occidentalis</i>	1	17.2	18.3	37.9	26.6	198
<i>Propithecus verreauxi</i>	2	18.9	17.1	35.4	28.6	172
<i>Indri indri</i>	2	15.2	18.5	43.3	23.0	141
<i>Daubentonia madagascariensis</i>	1	17.9	17.2	38.1	26.8	111
<i>Loris tardigradus</i>	2	19.0	14.4	42.0	24.6	135
<i>Nycticebus coucang</i>	2	16.3	16.5	39.5	27.7	157
<i>Perodicticus potto</i>	2	16.5	15.5	45.5	22.5	131
<i>Galago senegalensis</i>	2	14.9	15.6	37.4	32.1	203
<i>Otolemur crassicaudatus</i>	2	14.3	14.4	36.1	35.2	206
<i>Galagoides demidovii</i>	2	15.3	15.2	40.9	28.6	208
<i>Tarsius spp.</i>	4	9.87	15.2	43.7	31.4	227
Primates, Simians						
<i>Callithrix jacchus</i>	2	17.7	26.8	38.3	17.2	210
<i>Cebuella pygmaea</i>	2	18.8	26.5	37.0	17.7	258
<i>Saguinus midas</i>	2	20.3	20.8	36.1	22.8	165
<i>S. oedipus</i>	2	16.3	19.8	39.4	24.5	193
<i>Cebus albifrons</i>	1	22.4	14.8	38.3	24.5	75.0
<i>Aotus trivirgatus</i>	2	21.6	18.3	39.5	20.6	174
<i>Callicebus moloch</i>	2	18.6	16.1	39.6	25.7	184
<i>Saimiri sciureus</i>	2	15.1	21.3	38.8	24.8	130
<i>Pithecia monachus</i>	2	20.8	22.2	30.9	26.1	118
<i>Alouatta seniculus</i>	2	19.8	20.3	35.5	24.4	93.7
<i>Ateles geoffroyi</i>	1	14.5	13.7	47.9	23.9	49.2
<i>Lagothrix lagotricha</i>	2	17.1	13.1	43.2	26.6	55.6
<i>Macaca mulatta</i>	2	19.9	20.5	40.8	18.8	93.9
<i>Cercocebus albigena</i>	2	18.6	11.5	40.9	29.0	89.1
<i>Papio anubis</i>	2	13.5	9.80	45.0	31.7	56.0
<i>Cercopithecus ascanius</i>	2	15.8	11.3	39.9	33.0	82.0
<i>C. mitis</i>	1	18.6	13.0	38.2	30.2	114
<i>Miopithecus talapoin</i>	2	19.2	18.2	39.4	23.2	105
<i>Erythrocebus patas</i>	2	17.1	20.1	45.3	17.5	86.3
<i>Colobus badius</i>	2	15.2	17.5	42.0	25.3	77.6
<i>Pygathrix nemaeus</i>	1	22.4	19.5	29.0	29.1	86.2
<i>Nasalis larvatus</i>	1	14.6	17.2	39.0	29.2	92.9
<i>Hylobates lar</i>	1	25.1	9.30	41.2	24.4	60.9
<i>Pan troglodytes</i>	1	20.0	8.20	46.5	25.3	30.9
<i>Gorilla gorilla</i>	2	25.5	14.3	42.8	17.4	28.9
<i>Homo sapiens sapiens</i>	2	20.2	12.7	41.2	25.9	14.1

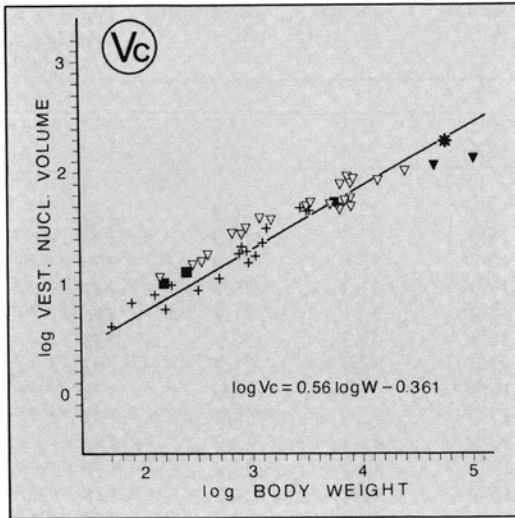


Fig. 1. The vestibular nuclear complex (V_c , mm^3) volumes plotted against body weight (W , g) on a double-logarithmic scale. The reference line, which is determined by least-square regression analysis, is placed through the average of prosimians. Squares = Scandentia; crosses = prosimians; open triangles = monkeys; black triangles = apes; star = man.

weight in a double-logarithmic scale, and regression lines were determined. In the present paper, the regression line of the 18 prosimians was used as the reference base line; its slope is 0.56. In the formula y is the value of each volume in cubic millimeters and x is the body weight in grams (example $\log y = 0.56 \log x - 1.142$). The distance from the reference line expresses the degree of deviation of the nuclear volume from those of a typical prosimian. Figure 1 shows the regression line for the volumes of the total vestibular complex.

The coefficients of correlation between the vestibular nuclear volumes and the volumes of other brain structures, and between their size indices were also calculated (table V). Volumes of the ventral pons and the cerebellar nuclei were taken from Matano et al.

Table IV. Size indices and ratios of vestibular nuclei

	N	Size indices				
		superior	lateral	medial	descending	complex
Tupaiidae	2	1.41	1.47	1.16	1.23	1.28
Cheirogaleidae	3	0.96	0.88	0.82	0.84	0.86
Lemurini	2	1.29	1.30	1.01	1.29	1.18
Lepilemurini	1	0.64	0.71	0.78	0.80	0.75
Indriidae	4	1.01	1.07	0.97	0.86	0.96
Daubentoniidae	1	1.38	1.30	1.27	1.21	1.27
Lorisinae	3	0.88	0.78	0.93	0.76	0.85
Galaginae	3	1.04	1.03	1.16	1.30	1.25
Tarsiidae	1	0.70	1.06	1.35	1.31	1.18
Prosimians		1.00	1.00	1.00	1.00	1.00
Callitrichidae	4	1.65	2.09	1.48	1.09	1.50
Cebidae	8	1.46	1.35	1.30	1.12	1.28
Cercopithecinae	7	1.40	1.18	1.40	1.19	1.30
Colobinae	3	1.00	1.01	0.90	0.94	0.95
Monkeys		1.41	1.39	1.31	1.11	1.28
Pongidae	3	1.05	0.44	0.83	0.61	0.74
Hominidae	1	1.00	0.61	0.88	0.75	0.82

[1985a, c] and those of the striatum from Stephan et al. [1981].

As additional data, the ratios of the volume of the vestibular complex to the total brain volume were calculated and multiplied by 10^5 . The percentage of the volume of each vestibular nucleus to the whole complex was also obtained and are listed in tables III and IV. Primates were classified according to Corbet and Hill [1980].

Results

Volumetric data of the vestibular nuclei are shown in table I.

The Superior Vestibular Nucleus

The regression line of the prosimians (18 species) is $\log y = 0.56 \log x - 1.142$.

High indices (1.41) are found in Tupaiidae, especially in *Tupaia glis* (1.58), the

highest of all prosimians investigated. Among prosimians from Madagascar, the values of *Lemur* (1.49) and *Daubentonia* (1.38) are high, and contrast with *Lepilemur* (0.64). Among Lorisidae, indices of *Galagoides* (1.20) are much higher than other species. Values of *Tarsius* (0.70) are the lowest among the four vestibular nuclei.

In anthropoids, the indices of the Callitrichidae are very high (1.65 on average), especially those of *Cebuella* (1.83). If *Alouatta*, *Ateles* and *Lagothrix* (average 0.87) are excluded from the Cebidae, the other species (*Cebus*, *Aotus*, *Callicebus*, *Saimiri*, *Pithecia*) show far higher indices (1.86). The superior nucleus is well developed in 6 species of Cercopithecinae (1.40 on average), especially in *Miopithecus* (1.99), whereas the values of *Papio* (0.69) are exceedingly low.

The lesser ape (*Hylobates*) has a remarkably high index of the superior nucleus (1.62) in contrast to the other three vestibular nuclear values. Poor development of this nucleus is found in *Gorilla* (0.74) and *Pan* (0.80), both values that are below those of man (1.00).

The percentage distribution of the superior nucleus to the whole complex ranges from about 15 to 25%. *Tarsius* has an exceptionally low value (9.9%).

The Lateral Vestibular Nucleus

The regression line for the lateral nucleus is $\log y = 0.56 \log x - 1.133$. The equation is very similar to that of the superior nucleus.

Indices in Scandentia, especially *Tupaia* (1.58) are high. The lateral nucleus indices of Madagascan Lemurini (1.30) and *Daubentonia* (1.30) are also high, in contrast to those of *Lepilemur* (0.71). *Loris* (0.65) and *Pero-dicticus* (0.70) are far lower than *Galagoides* (1.17) and *Tarsius* (1.06).

Ratios

superior %	lateral %	medial %	descend- ing, %	total brain × 100,000
18.0	19.4	35.2	27.4	325
18.8	17.2	36.3	27.7	205
18.0	18.6	32.7	30.7	144
14.2	16.0	39.7	30.1	208
17.3	18.8	38.9	25.0	188
17.9	17.2	38.1	26.8	111
17.3	15.4	42.4	24.9	141
14.8	15.0	38.1	32.1	206
9.90	15.1	43.7	31.3	227
16.7	16.9	38.5	28.0	180
18.3	23.5	37.7	20.5	206
18.7	17.4	39.3	24.6	110
17.5	19.9	41.3	26.3	89.5
17.4	18.1	36.7	27.8	85.6
18.1	17.8	39.3	24.8	118
23.5	10.6	43.5	22.4	40.2
20.2	12.7	41.2	25.9	14.1

Table V. Correlation coefficients between logarithms of volumes of striatum, ventral pons, three cerebellar nuclei and vestibular nuclei

Volumes	Size indices					
	striatum ^a	ventral pons	medial nucleus (cerebell.)	interpositus nucleus (cerebell.)	lateral nucleus (cerebell.)	vestibular nuclei
Striatum^a						
T		0.824	0.368	0.667	0.819	0.340
P	–	0.869	0.747	0.692	0.769	0.624
S		0.572	0.302	0.468	0.692	0.220
Ventral pons						
T	0.992		–0.006	0.499	0.906	0.036
P	0.989	–	0.668	0.768	0.738	0.689
S	0.993		–0.367	–0.001	0.874	–0.463
Medial nucleus (cerebellum)						
T	0.969	0.960		0.728	0.166	0.830
P	0.979	0.975	–	0.745	0.696	0.703
S	0.943	0.945		0.813	–0.019	0.884
Interpositus nucleus (cerebellum)						
T	0.971	0.964	0.990		0.608	0.661
P	0.976	0.975	0.991	–	0.737	0.568
S	0.949	0.948	0.980		0.300	0.708
Lateral nucleus (cerebellum)						
T	0.988	0.990	0.969	0.971		0.159
P	0.980	0.973	0.981	0.979	–	0.610
S	0.989	0.994	0.959	0.961		–0.188
Vestibular nuclei						
T	0.970 (1.5)	0.969 (1.9)	0.981 (1.2)	0.983 (1.4)	0.973 (1.6)	
P	0.970 (1.2)	0.977 (1.4)	0.981 (1.2)	0.978 (1.4)	0.977 (1.3)	–
S	0.957 (1.6)	0.966 (2.1)	0.968 (1.2)	0.969 (1.5)	0.968 (1.9)	

Numbers in brackets give the slope for the vestibular nuclei. T = Total species investigated; P = prosimians; S = simians

^a Without pallidum.

Among anthropoids, two species of marmosets have the highest lateral nucleus indices (2.36 in *Callithrix* and 2.51 in *Cebuella*). These values are also far higher than the other three vestibular nuclei indices. As with the superior, the lateral nucleus is well devel-

oped in 5 species of Cebidae (average 1.70 in *Aotus*, *Callicebus*, *Saimiri*, *Cebus* and *Pithecia*), but not in *Ateles*, *Alouatta* and *Lagothrix* (0.77 on average). In the Cercopithecinae, a relatively wide range of indices is found, from *Papio* (0.49) to *Miopithecus* (1.86),

and the average values (1.18) are higher than those of Colobinae (1.01). The lowest indices through all materials investigated are found in *Pan* (0.32) and *Gorilla* (0.42). The lateral nucleus values in man (0.61) are the lowest of the 4 vestibular nuclei.

Prosimians and Scandentia have a relatively narrow range of the percentage ratio of the lateral nucleus to the whole complex (14.2–21.2%). A distinctly higher percentage (26.7%) is found in marmosets. On the other hand, ape and human lateral nucleus percentages are the lowest of the vestibular nuclei (10.6% in Pongidae, 12.7% in man).

The Medial Vestibular Nucleus

The regression line for the medial nucleus in prosimians is $\log y = 0.56 \log x - 0.778$.

The index of *Lemur* is relatively low compared to the other three nuclei. As with the above mentioned nuclei, indices of the medial nucleus of *Galagoides* (1.39) and *Tarsius* (1.35) are distinctly higher (0.93 in average) than in Lorisinae.

Callitrichidae also have very high medial indices (1.48 on average). Five species of cebus monkeys (*Cebus*, *Aotus*, *Callicebus*, *Saimiri* and *Pithecia*; average 1.56) have a high index, with *Saimiri* having the highest in this study (1.81). Except for *Papio* (0.99), an above average development of the medial nucleus is found in Cercopithecinae (1.47 on the average). The indices of Colobinae (0.90) are lower than the prosimian average. The value for *Gorilla* (0.54) is the lowest of all specimens investigated.

The distribution percentages of the medial nucleus in relation to the whole complex show that the medial is the largest nucleus; it occupies between 29 and 46.5% of the total complex in all primate species (from 33 to 44% when averaged by family).

Descending Vestibular Nucleus

The regression line of the descending nucleus in the prosimians is $\log y = 0.56 \log x - 0.910$.

The indices in prosimian species show the same tendency as those of the other three nuclei. The indices of the Galaginae (1.30) and Tarsiidae (1.31) are clearly higher than those of the Lorisinae (0.76). Indices of the marmosets (0.96) are much lower compared with those of the other vestibular nuclei. As was the case with the other three nuclei, three species of Cebidae show low indices (*Alouatta*, *Ateles*, *Lagothrix*; 0.75 on average) in contrast to the higher indices of *Cebus*, *Aotus*, *Callicebus*, *Saimiri*, *Pithecia* (1.34 on average). The index of *Saimiri* (1.57) is the highest of all specimens investigated. Values of *Macaca* (0.83) and *Erythrocebus* (0.84) are much lower compared with the values of the other three nuclei. The index of *Gorilla* (0.30) is the lowest among all vestibular nuclear indices. Values of both man (0.75) and apes (0.61 on average) are below the prosimian average.

Distribution percentages of the descending nucleus within the whole complex are relatively constant. Three quarters of all species including man fall between 20 and 30%.

Vestibular Nuclei Complex

The regression line of the vestibular complex in prosimians is $\log y = 0.56 \log x - 0.361$.

Scandentia have remarkably high indices (1.28 on average). Among Madagascan prosimians, the indices of Lemurini (1.18) and *Daubentonina* (1.27) are above the prosimian average, while Indriidae (0.96), *Lepilemur* (0.75) and Cheirogaleidae (0.86) are lower. Among Lorisidae, a distinct difference is

seen between the indices of Lorissinae (0.85) and Galaginae (1.25). The index of *Tarsius* is above the average (1.18).

In anthropoids, 4 species of Callitrichidae show very high indices (1.50 on average). Three species of Cebidae have remarkably low indices (*Alouatta*, *Ateles* and *Lagothrix*; 0.84 on average). Another 5 species, on the other hand, show a good development of the vestibular complex (*Aotus*, *Callicebus*, *Saimiri*, *Pithecia* and *Cebus*; 1.54 on average). The highest values are found in *Saimiri* (1.78). The whole-complex indices in Cercopithecinae are above the prosimian average, with the exception of *Papio* (0.85).

The index of *Hylobates* is between the average values of Cercopithecinae and Colobinae. The index of *Homo* (0.82) is higher than those of the great apes, which show the lowest values (0.66 in *Pan*, 0.48 in *Gorilla*).

Tables III and IV show the ratios of the vestibular complex volume to total brain volume. The highest values are found in Scandentia (325 on average). Pongidae and human ratios are inversely proportional to their size indices (pongids: 22.9 in ratios, 0.57 in size indices; humans: 14.1 in ratios, 0.82 in size indices).

Correlation coefficients and slopes describing the volume and size indices of the vestibular complex with other motor structures are shown in table V. All coefficients for volumes are highly correlated. In anthropoids, the ventral pons and the lateral cerebellar nucleus increase in size much more than the vestibular complex (slope of the ventral pons = 2.1, of the lateral cerebellar nucleus = 1.9). The size indices in all species clearly correlate with the medial cerebellar nucleus ($r = 0.830$, d.f. = 44; $p < 0.01$). On the other hand, lower coefficients are found with the striatum ($r = 0.340$, d.f. = 44; $p <$

0.05), and no correlation is found with the ventral pons ($r = 0.036$) or with the lateral cerebellar nucleus ($r = 0.159$). In anthropoids, the ventral pons has a negative correlation with the vestibular nuclei ($r = -0.463$, d.f. = 24; $p < 0.05$).

Discussion

The size indices of the vestibular complex contrast remarkably with those of the neocortex, the ventral pons or the lateral cerebellar nucleus [Matano et al., 1985a, c]. That is, the phylogenetic development of the vestibular complex is independent of the trends seen in the neocortex or the ventral pons (fig. 1). Its development may be a structural specialization reflecting species-specific locomotor behavior. Inputs from the vestibular nuclei are thought to be integrated into the cerebellar control of posture and simple automatic movements [Eccles, 1982].

As is well known, *Tupaia* employs a rapid jerky scurrying movement on the ground or in trees during daytime [Napier and Napier, 1967]. The size indices show *Tupaia* to have relatively enlarged vestibular nuclei. The development is better than that of the cerebellar nuclei, ventral pons or striatum. This suggests that the normal locomotor pattern of *Tupaia* is largely dependent on the simpler automatic movements [Eccles, 1982].

Among Galaginae, the locomotor type of *Otolemur* lies somewhere between those of *Galago* and *Perodicticus* [Bearder and Doyle, 1974]. But the locomotion of *Galago* and *Galagoides*, as well as *Tarsius*, is classified as vertical clinging and leaping (VCL) [Napier and Napier, 1967]. These three typical VCL species have higher than average vestibular complex indices (1.20), and these values are

higher than those in three species of Lorisiinae whose locomotion is classified as slow climbing quadrupedalism (SCQ; 0.85 on average for *Nycticebus*, *Perodicticus* and *Loris*; $p < 0.05$). The allometric variation in the vestibular nuclei between these two groups may reflect the size difference of the semicircular canals in these animals. Morphological [Gray, 1951; Hadziselimovic and Savkovic, 1964; Curthoys et al., 1976] and physiological evidence [Blanks et al., 1975] suggests that fast-moving creatures demand a larger curvature in the semicircular canals, because sensitivity of this organ increases proportionately to the size of the curvature. Jones and Spells [1963] and Jones [1974] measured the dimensions of the semicircular canals in more than 80 vertebrate genera. The allometric values of the radii of semicircular canal curvatures to body weight in *Tarsius* were relatively large, near to the range of birds, whereas those of *Nycticebus* were relatively small and within the range of mammals [Matano et al., 1985b, 1986].

On the other hand, the two locomotor groups have very similar ventral pontine indices (1.01 on the average for *Tarsius*, *Galago*, *Galagoides* and 1.02 for *Loris*, *Nycticebus*, *Perodicticus*). This result suggests that the voluntary movement is weak in the unique leaping behavior of VCL type (*Tarsius*, *Galago*, *Galagoides*).

The locomotion of *Lepilemur* and Indriidae is also classified by Napier and Napier [1967] as VCL. But their indices of the vestibular complex are low, as are those of the ventral pons. They are much larger in body weight compared with *Galago* and *Tarsius*. The differences suggest that the pattern and biomechanical processes of VCL may differ between the small- and the large-bodied groups.

The vestibular nuclear indices of *Lemur* (1.24) and *Daubentonia* (1.27) are above the prosimian average. These two species also show the highest ventral pontine indices among prosimians (1.61 and 2.26, respectively). In addition to including the pyramidal tract, the ventral pons is an important intermediate station in the neocortex-neocerebellar circuit for planning, programming and performing voluntary movements [Eccles, 1982; Matano et al., 1985a, c]. From these results, both species are considered to be between prosimians and monkeys, not only in abilities of performing the simpler automatic movements but also complex voluntary ones [Eccles, 1982]. The high indices of the ventral pons in *Daubentonia* may be related to the great skill of using the skeletal finger to capture larvae of wood-boring insects.

Jones and Spells [1963] and Jones [1974] measured the semicircular canal size (radii of curvature, R) of several anthropoids: common marmoset, black ape, green monkey, hocheur monkey, yellow baboon and man. Although there is some variation in body mass, generally speaking, the anthropoid values of log R per body mass are above the regression line of the 51 mammalian species. The common marmoset has higher values than the yellow baboon. This finding corroborates our findings on the vestibular complex indices of both species.

Among the arboreal quadrupedals of New World monkeys [NAQ; Napier and Napier, 1967], the springers (*Callitrichidae*, *Aotus*, *Callicebus*) and the climbers (*Cebus*, *Saimiri*, *Pithecia*) overlap in the vestibular complex indices, but the New World semibrachiators (NSB: *Alouatta*, *Ateles*, *Pygathrix*) show remarkably low indices when compared with the former two locomotor groups (the

former two: 1.52, vs. NSB: 0.84). On the other hand, the ventral pontine indices are clearly bigger in climbers than in springers (3.25 vs. 1.47 on average). The semibrachiators have ventral pontine indices within the range of those of climbers and springers.

The New World monkey *Cebus albifrons* occupies a conspicuous position. As already reported [Matano et al., 1985a, c], *Cebus* occupies a very high position in the size indices of the ventral pons (4.72) and the lateral cerebellar nucleus (2.98). This may be closely related to its agile locomotion and considerable manipulative skill in handling objects [Napier and Napier, 1967]. Also, the vestibular complex index of *Cebus* is relatively high (1.28). This trend seen in the size indices suggests that *Cebus* excels not only in the skilled movement, but also in the simple automatic movement.

The vestibular complex indices in Old World arboreal quadrupeds (OAQ; *Cercopithecus*, *Cercopithecus*, *Miopithecus*) completely overlap with those of NAQ. On the other hand, Old World terrestrial quadrupeds (OTQ; *Macaca*, *Papio*, *Erythrocebus*) have lower indices compared with OAQ. As well as the case of NSB, relatively poor development is seen in the Old World semibrachiators (OSB; Colobinae) compared with those of other two types of locomotion (fig. 2; average 0.95 in OSB, 1.42 in OAQ, 1.15 in OTQ). The ventral pontine indices of OSB are also relatively low and within the lower portion of the range of the other two (OAQ, OTQ) types (fig. 3).

The vestibular complex indices in great apes (*Pan*, 0.66, and *Gorilla*, 0.48) are the lowest of all investigated species. The true brachiator, *Hylobates*, however, has indices suggesting both simple automatic and voluntary skillful movements (vestibular complex:

1.07, ventral pons: 5.38, lateral cerebellar nucleus: 2.81). These indices resemble those of *C. albifrons*.

Human indices of the vestibular complex are low, though above those of the great apes. The vestibular values in man and the great apes correspond well with the indices of the medial cerebellar nucleus [Matano et al., 1985a]. This correspondence supports a close functional relationship between both nuclei as discussed later. In any event, finely organized, complex voluntary movements are extremely conspicuous in man, and the behavioral quality is associated with the development of the ventral pons and the lateral cerebellar nucleus [Matano et al., 1985a, c]. The development of the vestibular nuclei does not seem to be necessary for human bipedalism.

Brodal [1984] said that, viewed from topography, cytoarchitecture and fiber connections of the vestibular nuclear complex, it is difficult to draw reliable conclusions about phylogenetic trends of the vestibular nuclear complex, but that the lateral vestibular nucleus differs by species in topography and cytoarchitecture. According to Brodal, giant cells in the cat are relatively more numerous and in general larger in the caudal part of the lateral nucleus than in the rostral part. In *Macaca*, the differences between the rostral and caudal regions are more distinct than in the cat. But in man, one can fully distinguish between the lateral and medial parts. The medial part, situated ventrorostrally, and the lateral part, situated more dorsocaudally, show differences resembling those between the rostral and caudal regions, respectively, of the lateral nucleus in *Macaca*. Thus, Gerhard and Olszewski [1969] observed that only a subdivision into lateral and medial parts is justified in apes and man. A more

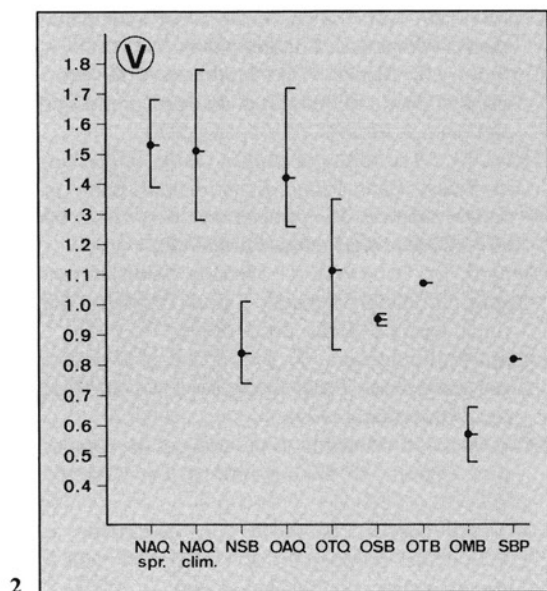
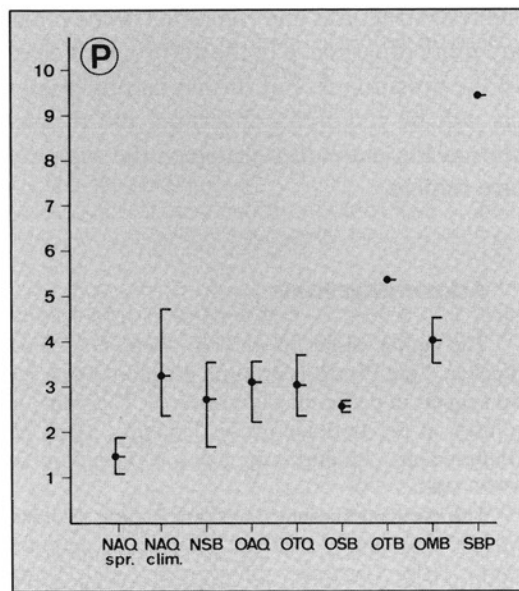


Fig. 2. The size indices (means and range) of the vestibular nuclear complex in anthropoids. NAQ springers (spr.; *Callithrix*, *Cebuella*, *Saguinus midas*, *S. oedipus*, *Aotus*, *Callicebus*), NAQ climbers (clim.; *Cebus*, *Saimiri*, *Pithecia*), NSB (*Alouatta*, *Ateles*, *Lagothrix*), OAQ: (*Cercocebus*, *Cercopithecus ascanius*, *C. mitis*, *Miopithecus*), OTQ (*Macaca*, *Papio*, *Ery-*



throcebus), OSB (*Pygathrix*, *Colobus*, *Nasalis*), OTB (Old World true brachiators, *Hylobates*), OMB (Old World modified brachiators, *Pan*, *Gorilla*) and SBP (specialized bipedal hominid, *Homo sapiens*) are compared.

Fig. 3. The size indices of the ventral pons in anthropoids. Abbreviations as in figure 2.

detailed relation between our results on the lateral nucleus by allometry, which shows poor development in apes and man, and the topographical and cytoarchitectural results by Brodal [1984] and Gerhard and Olzowski [1969] in apes and man remains unexplained at present.

Finally, we refer to the relation between the vestibular complex and the cerebellar nuclei. Among Chiroptera [Baron, 1978], insectivores, prosimians and anthropoids [Matano et al., 1985a], the medial cerebellar nucleus is the least progressive and the lateral cerebellar nucleus the most progressive. The medial cerebellar nuclear size indices resem-

ble that of the vestibular complex. Correlations between the complex and the medial and lateral cerebellar nuclear size indices demonstrate this trend clearly (table V). These correlation coefficients reflect the close relationship in structure and function between the vestibular complex and the medial cerebellar nucleus, as well as the mutual weak relation in the structure and function between the whole complex and the lateral cerebellar nucleus, the ventral pons (0.036) and the striatum (0.340) [Stephan, 1979]. These trends are more remarkable in the anthropoids than in the prosimians, which suggests that differentiation and develop-

ment of structures and functions in the brain are more progressive in the anthropoids than in the prosimians. But further careful analysis will be required concerning the discussion of the correlation between the separate size indices.

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